Anglo-Chinese Junior College

Physics Preliminary Examination
Higher 2

A Methodist Institution
(Founded 1886)

CANDIDATE NAME		CLASS	

CENTRE NUMBER	S	3	0	0	4	INDEX NUMBER				

PHYSICS **9478/03**

Paper 3 Longer Structured Questions 2 September 2026
2 hours

You must answer on the question paper.

No Additional Materials are needed.

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and index number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiners' use only	
Section A	
1	/ 5
2	/ 12
3	/ 12
4	/ 14
5	/ 12
Total	/ 55
Section B	
6 / 7	/ 20
Grand Total	/ 75

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on / by a gas	W	=	$p\,\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{Gm}{r}$
temperature	T / K	=	T / °C + 273.15
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$

mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{(x_0^2 - x^2)}$
electric current	I	=	$nAvq$
resistors in series	R	=	$R_1 + R_2 + \cdots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \cdots$
energy in a capacitor	U	=	$\frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	Q	=	$Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	Q	=	$Q_0 e^{-\frac{t}{\tau}}$
RC time constant	τ	=	RC
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 N I}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 n I$
energy states for quantum particle in a box	E_n	=	$\frac{h^2}{8mL^2}n^2$
radioactive decay	x	=	$x_0 e^{-\lambda t}$
radioactive decay constant	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in the spaces provided.

1 The viscosity of a fluid η is determined by the equation

$$\eta = \frac{\pi P r^4}{8l\left(\frac{V}{t}\right)}$$

where $\frac{V}{t}$ is the volume rate of flow of the fluid through a horizontal pipe,
l is the length of the horizontal pipe,
r is the internal radius of the pipe,
P is the pressure difference across the pipe.

(a) Show that when expressed in base SI unit, the units for η is kg m^{-1} s^{-1}.

[2]

(b) In an experiment to determine η for water, a student quoted his result as 1.137×10^{-3} kg m^{-1} s^{-1}.

He estimated the percentage uncertainties in his measurement of $\frac{V}{t}$, l, r, P as $\pm 3\%$, $\pm 2\%$, $\pm 0.5\%$ and $\pm 5\%$ respectively.

(i) Calculate the percentage uncertainty in the value of η.

percentage uncertainty = …… % [2]

(ii) Hence, express the value of η together with its absolute uncertainty. Give your answer to an appropriate number of significant figures.

η = (…… ± ……) × 10^{-3} kg m^{-1} s^{-1} [1]

[Total: 5]

2 **(a)** State what is meant by *electric field strength*.

……

……

……

…..…… [2]

(b) In a crystal of sodium chloride, a sodium ion of charge +*e* is separated from its closest chloride ion of charge –*e* by a distance of 2.8×10^{-10} m.

Assume that the ions are point charges.

(i) Calculate the magnitude of the electric field strength that one sodium ion creates at its closest chloride ion.

electric field strength = …….. N C^{-1} [2]

(ii) Calculate the attractive force between the two ions.

force = …… N [1]

(c) The force in **2(b)(ii)** does not accelerate the sodium and chloride ions towards each other due to presence of other ions.

Figure 2.1 represents a plane of sodium and chloride ions at their equilibrium distance.

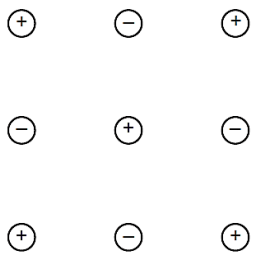


Figure 2.1

(i) With reference to Figure 2.1, state and explain the resultant force on the central ion.

……

……

……

……

……

……. [3]

(ii) The central ion vibrates out of position. State the effect of the resultant force on the ion.

……. [1]

(iii) Suggest what affects the equilibrium distance between any two adjacent sodium and chloride ions.

……

……. [1]

(d) Sodium chloride and potassium chloride are both essential mineral salts that act as vital electrolytes in the human body. They have similar physical properties and potassium chloride is often used as a sodium-free salt substitute to help lower blood pressure.

The equilibrium distance for sodium chloride and potassium chloride is 0.28 nm and 0.31 nm respectively.

Suggest, with a reason, which of the two salts has a higher melting point.

……

……

……

……

…… [2]

[Total: 12]

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3 (a) The equation $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$ can be used in calculations to determine the

internal energy of a gas.

(i) State the meaning of the following:

1. m .....

2. $\langle c^2 \rangle$ .....

.....

3. $\frac{1}{2}m\langle c^2 \rangle$ .....

.....

[3]

(ii) Calculate the internal energy of one mole of an ideal gas at 27.0 °C.

internal energy = ... J [2]

(b) The atmosphere is mostly a mixture of nitrogen and oxygen molecules.

For molecules in the atmosphere, deduce the ratio

$$\frac{\text{r.m.s. speed of a nitrogen molecule}}{\text{r.m.s. speed of an oxygen molecule}}$$

.

Mass of a nitrogen molecule = 4.65×10^{-26} kg.
Mass of an oxygen molecule = 5.31×10^{-26} kg.

ratio = ... [2]

(c) During the first part of the power stroke in a diesel engine, the pressure of the burning gas remains constant at 8.00×10^6 Pa, while the volume of the gas

increases from 3.00×10^{-5} m^3 to 7.10×10^{-5} m^3.

(i) Calculate the work done by the gas during the first part of the power stroke.

work done = …… J [2]

(ii) While this expansion is taking place, the temperature of the gas rises by 900 K. The mass of the gas is 1.27×10^{-3} kg and its specific heat capacity at constant pressure is 1.01×10^{3} J kg^{-1} K^{-1}.

Calculate the heat gained by the burning gas.

heat gained = …… J [2]

(iii) Deduce, using your answers to **3(c)(i)** and **3(c)(ii)**, the change in the internal energy of the burning gas.

change in internal energy = …… J [1]

[Total: 12]

4 **(a)** State **two** conditions required for two-source interference pattern to be observed.

1. ……..……

……..……

2.
……

……....
[2]

(b) A structural engineer is using a laser-based monitoring system to detect microscopic shifts in a bridge's support beam.

A Young's Double Slit experiment where a monochromatic laser using red light of wavelength 633 nm was shone through two narrow slits attached to the beam. The resulting interference pattern is projected onto a screen located 2.50 m away.

At the start of the day, the distance between the centre of the central maximum and the third order maximum is measured to be 1.85 cm.

(i) Determine the separation between the two slits.

separation = …… mm [3]

(ii) Later in the afternoon, due to thermal expansion, the slits physically move slightly further apart.

State and explain any changes to the **distance between the bright fringes** on the screen.

……..……..……

……..……

……

……..... [2]

(iii) As the light travels from the slits to the screen, it passes through regions of varying temperatures, which affects the refractive index of the air.

The refractive index n of air is given by the expression

$$n = \frac{c}{v}$$

where c is the speed of light in vacuum and v the speed of light in air.

Suggest how the fluctuation in n affects the interference pattern formed on the screen.

……..……

……..……

……

……..... [2]

(c) Khongoryn Els ~~of the Gobi Desert is known as the Singing Dunes~~.

At the Khongoryn Els, a specific frequency is produced when a surface layer of dry, "boiling" sand slides over a more compact, stationary layer during an “avalanche”. This gap between the moving surface and the stationary bed (the vibrating layer) acts like a pipe **closed at one end** (the bottom) and **open at the top**. A 3-D schematic diagram of the layers is shown in Figure 4.1

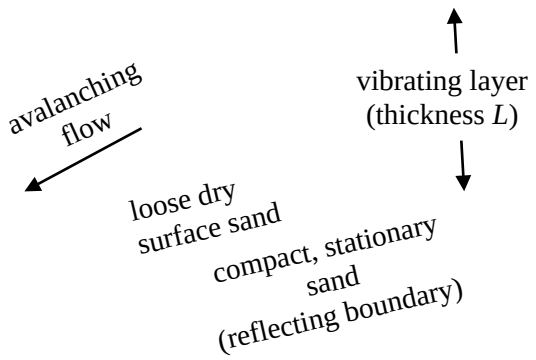


Figure 4.1

The thickness of this resonating dry sand layer L is 1.20 m and the speed of sound in the hot, sandy air is 348 m s^{-1}.

(i) The vibrating dry sand layer of the singing dunes can be represented as a pipe of length 1.20 m as shown in Figure 4.2.

When the fundamental frequency is heard within this dry sand layer, on Figure 4.2,

1. label the positions of the pressure antinode **PA** and displacement antinode **DA**,

2. sketch the pressure variation of the standing wave pattern formed.

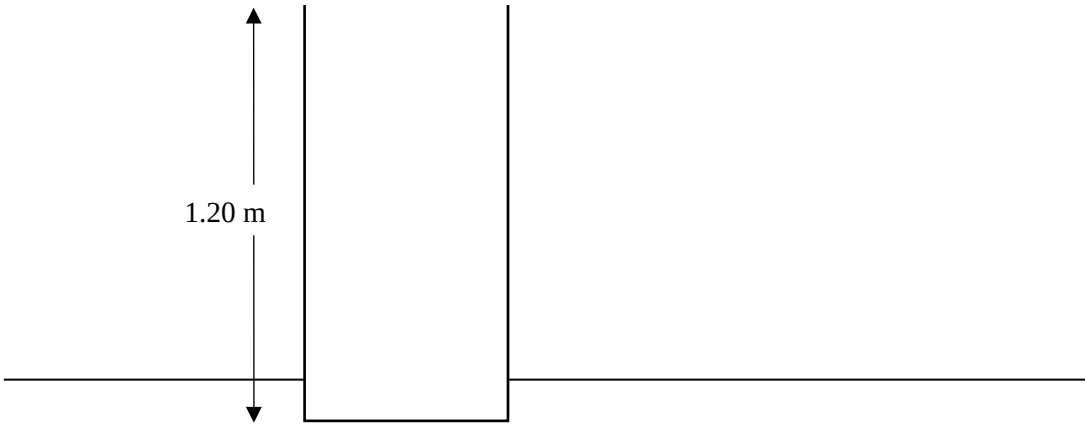


Figure 4.2

[2]

(ii) Calculate the fundamental frequency (the "hum") produced by the dune.

frequency = …… Hz [1]

(iii) In addition to the fundamental frequency, the dune also produces higher-

frequency overtones that contribute to its "eerie" sound.

Calculate the frequency of the next higher harmonic that could be produced within this 1.20 m sand layer.

frequency = …… Hz [2]

[Total: 14]

5	**(a)**	State the relationship between the energy *E* and momentum *p* of a photon.

			……..…… [1]
	(b)	An electron in a quantum dot is modelled as a particle in a one-dimensional infinite square well of width 3.0 nm. It makes a transition from quantum state n = 8 to n = 2 and emits a photon. An array of identical quantum dots produces many such photons, which are incident on a metal plate of work function 2.30 eV.	
		(i)	Calculate the threshold frequency of the metal.
			threshold frequency = …… Hz [2]
		(ii)	Show that the energy of the photon is 2.51 eV.
			[3]

		(iii)	Hence, determine the de Broglie wavelength of the fastest photoelectrons.
			de Broglie wavelength = …… m [3]
		(iv)	The photoelectrons are sent through a velocity selector before being directed at a double-slit forming an interference pattern on a screen. State and explain the effect on the interference pattern when the velocity selector is removed.
			…….. …….. …….. …….. …….. …….. [3]
			[Total: 12]

Section B

Answer **one** question in this section in the spaces provided.

6 **(a)** State the conditions necessary for an object to have *simple harmonic motion.*

……..……..

……..……..

……..……..

…… [2]

(b) A long bar magnet hangs from one end of a spring as shown in Figure 6.1.

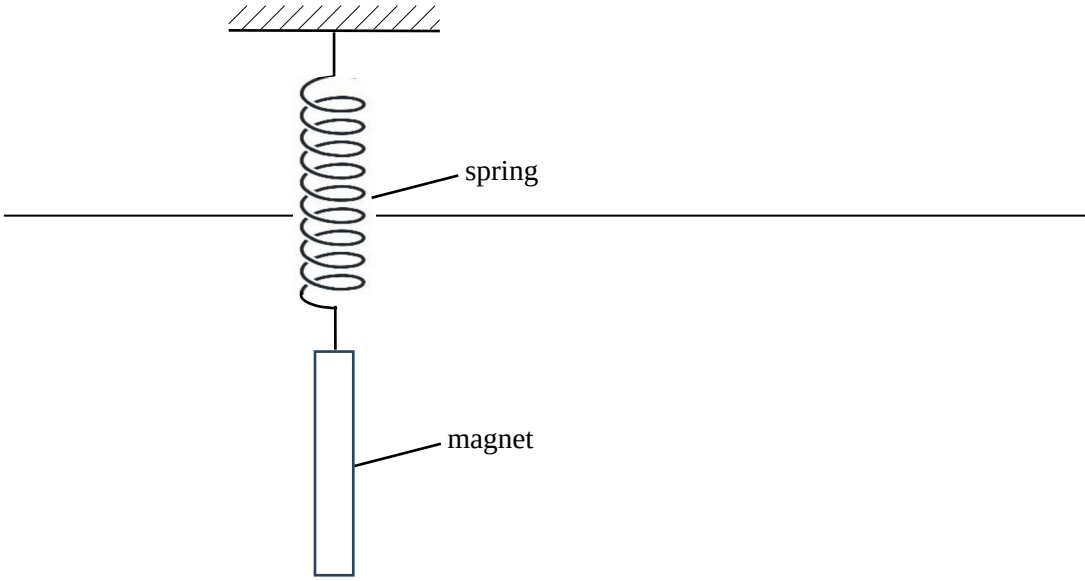


Figure 6.1

The magnet is displaced vertically downwards and then released. The subsequent vertical displacement x is found to vary with time as shown in Figure 6.2.

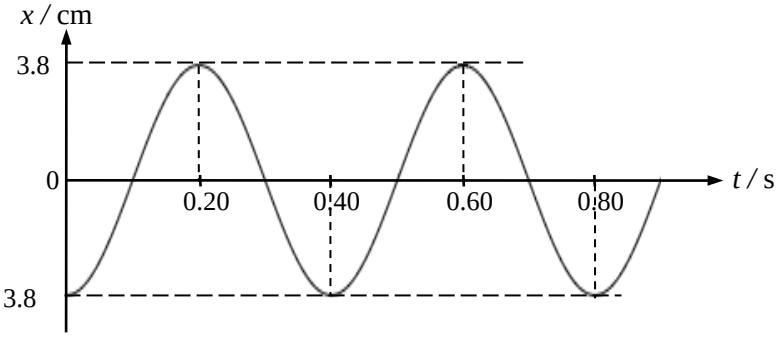


Figure 6.2

(i) Using values from Figure 6.2,

1. calculate the angular frequency ω of the oscillation of the magnet.

ω = ... rad s^{-1} [2]

2. determine the acceleration of the magnet when the displacement is zero and when the displacement is at its maximum.

acceleration when displacement is zero = ... m s^{-2}

acceleration when displacement is maximum = ... m s^{-2}
[3]

(ii) On Figure 6.3, sketch the graph of the variation with displacement of the velocity of the magnet. Label the axes with appropriate values.

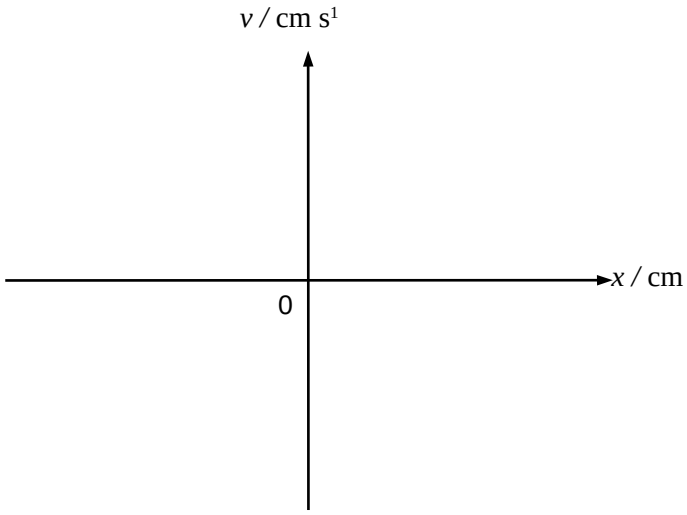


Figure 6.3

[2]

[Turn over

(iii) Determine the distance of the magnet from the equilibrium position when its speed is half of the maximum speed.

distance = …… cm [3]

(c) One pole of the magnet is now placed inside a coil of wire as shown in Figure 6.4. The magnet is given a small displacement and is then released so that the north pole of the magnet oscillates within the coil.

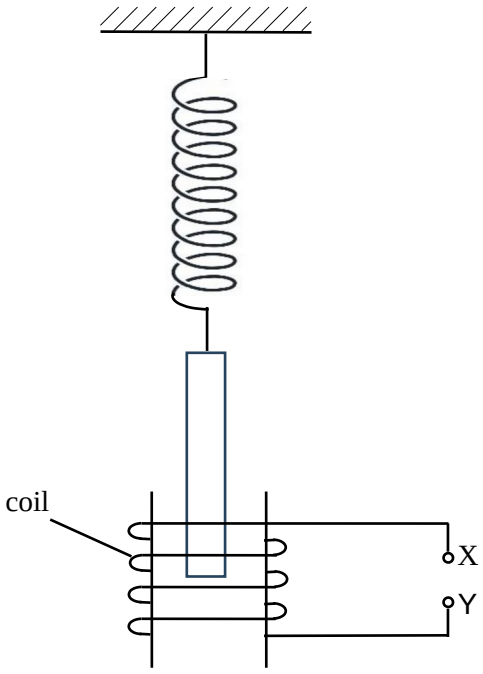


Figure 6.4

An induced e.m.f. is measured across the terminals XY of the coil.

(i) On Figure 6.5, sketch the variation with time of the induced e.m.f. as the magnet oscillates according to the displacement-time graph of Figure 6.2.

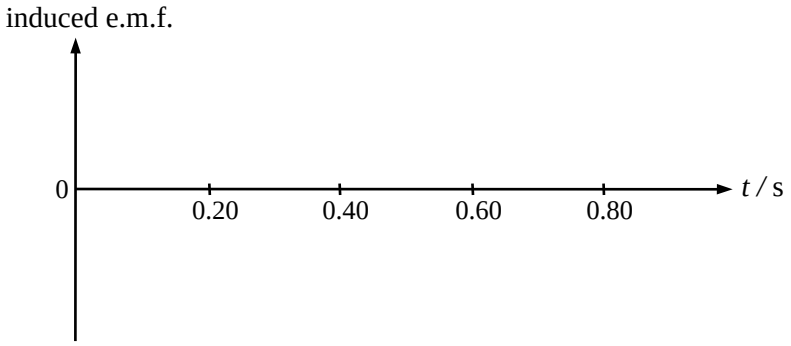


Figure 6.5

[2]

(ii) Using the principles of electromagnetic induction and with reference to Figure 6.2, explain the variation of the induced e.m.f. drawn in 6(c)(i).

……

……

……

……

……

…….. [3]

(iii) State and explain the change, if any, that occurs in the amplitude of the oscillations of the magnet when a resistor is connected across XY.

……

……

……

……

……

……

…….. [3]

[Total: 20]

7 Molybdenum-99 ^{99}Mo is a key radioisotope used to produce Technetium-99m ^{99m}Tc, one of the most widely used tracer in medical imaging like bone and heart scans. In a nuclear reactor, Molybdenum-99 is produced as a fission product when Uranium-235 ^{235}U absorbs a slow neutron.

(a) A typical fission reaction producing Molybdenum-99 is given by the equation:

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{99}_{42}Mo + ^{134}_{z}Sn + x\,^{1}_{0}n$$

(i) Determine the values of the proton number z and the integer x representing the number of emitted neutrons.

z = ……

x = ……

[2]

(ii) The binding energies per nucleon for the relevant nuclides are given below:

^{235}U : 7.59 MeV
^{99}Mo : 8.61 MeV
^{134}Sn : 8.45 MeV

Calculate the energy released in this fission reaction.

energy released = …… MeV [2]

(iii) This nuclear fission reaction is also called a *chain* reaction.

Suggest what is meant by a *chain* reaction.

……

…… [1]

(b) Once Molybdenum-99 is produced, it undergoes β^- decay to produce Technetium-99m ^{99m}Tc. Because ^{99m}Tc has a short half-life, it cannot be stored long-term. Instead, it is produced in hospitals using a portable generator containing Molybdenum-99.

The following data is provided:

Mass of Molybdenum-99 ^{99}Mo	: 98.90771 *u*
Mass of Technetium-99m ^{99m}Tc	: 98.90641 *u*
Mass of an electron	: 0.00055 *u*

(i) Radioactive decay is a *random* and *spontaneous* process.

Explain what the following terms mean.

1. *random*

……

…… [1]

2. *spontaneous*

……

…… [1]

(ii) Write down a balanced nuclear equation representing the β^- decay of Molybdenum-99 to Technetium-99m, including any fundamental particles emitted.

[1]

(iii) Determine the energy released in MeV during the decay of Molybdenum-99.

energy released = …… MeV [3]

(c) The half-life of Molybdenum-99 and Technetium-99m is 66 hours and 6.0 hours respectively.

(i) Define *decay constant*.

……

……... [1]

(ii) A fresh medical generator arrives at a hospital containing a sample of pure Molybdenum-99 with an initial activity of 12.0 GBq.

Determine the initial mass of Molybdenum-99 nuclei present.

initial mass = …… kg [3]

(iii) Hence determine the mass of Molybdenum-99 nuclei present after 30 days.

mass = …… kg [2]

(d) Technetium-99m is attached to a carrier molecule that targets a specific part of the body when injected into a patient. When Technetium-99m transitions to its ground state Technetium-99 ^{99}Tc, it emits a γ-ray photon. Technetium-99 has a much longer half-life of 2.11×10^5 years.

(i) Explain why Technetium-99m is highly suitable as a medical tracer.

……

…… [1]

(ii) Explain why Technetium-99 poses negligible radiation risk to the patient post-scan.

……

…… [1]

(iii) In contrast to medical tracers, industrial tracers that track fluid flow or leaks in deep underground chemical pipelines utilise radioisotopes with longer half-lives.

Suggest why a longer half-life is necessary for industrial pipeline applications.

……

…… [1]

[Total: 20]

End of Paper